

Excellent. At this point, **the README is part of your product**, not just documentation. A judge may spend 30–60 seconds on it before deciding whether to dive deeper.

README.md

Markdown



🎵 RagaChakra

AI Ritual Companion for Indian Classical Music

> Transform your mood into a mindful listening ritual using AI, Hindustani Classical Music

![Status] (https://img.shields.io/badge/Status-Hackathon-blue)

![React] (https://img.shields.io/badge/React-18-61DAFB)

![Node] (https://img.shields.io/badge/Node.js-20-green)

![MongoDB] (https://img.shields.io/badge/MongoDB-Atlas-green)

![Gemini] (https://img.shields.io/badge/AI-Gemini%202.5-orange)

✨ What is RagaChakra?

RagaChakra is an **AI Ritual Companion** that helps people discover the right Hindustani

- 🕒 Current time (Prahar)
- 😊 Emotional state
- 🧠 MBTI personality (optional)
- 📍 Local sunrise/sunset
- 📖 Traditional musicology

Unlike conventional music recommendation systems that rely on popularity or listening his

🎯 Problem

Today's recommendation engines optimize for engagement.

They recommend:

- Trending songs
- Viral playlists
- Endless scrolling

They do **not** consider:

- Time of day

- Emotional well-being
- Traditional musical knowledge
- Intentional listening

Indian Classical Music contains centuries of knowledge about which ragas are appropriate

That knowledge remains largely inaccessible to modern audiences.

💡 Our Solution

RagaChakra transforms recommendation into a guided ritual.

Instead of:

...

Song → Next Song

...

Users experience:

...

Mood

↓

AI Understanding

↓

Time-aware Raga

↓

Guided Ritual

↓

Reflection

↓

Memory

...

The AI doesn't simply recommend music.

It guides an experience.

🧠 Hybrid AI Architecture

Unlike traditional AI-only systems, RagaChakra uses a Hybrid AI Pipeline.

...

User Context

(Mood + Time + Location)

↓

Rule Engine

(Current Prahar)

↓

MongoDB

(Culturally Correct Ragas)

↓

Gemini 2.5 Flash

↓

Personalized Explanation

↓

Guided Ritual

...

This architecture ensures:

- Cultural correctness
- Explainable AI
- Low hallucination risk
- Reliable recommendations

🚀 Features

🌅 Time-aware Recommendation

Uses Prahar and SunCalc to identify the appropriate raga for the user's current time.

😊 Mood Check-in

Simple emotional onboarding.

Examples:

- Calm
- Anxious
- Focused
- Curious
- Tired

🤖 AI Thinking

Visible reasoning process.

...

- ✓ Understanding your mood
- ✓ Reading today's Prahar
- ✓ Comparing evening ragas
- ✓ Preparing your ritual

...

🎵 Recommendation Experience

Each recommendation includes:

- Confidence score
- Explainability card
- Embedded music
- Alternative recommendation
- Tomorrow preview

🧘 Ritual Mode

Guided listening experience.

Stage 1

Close your eyes

Stage 2

Listen

Stage 3

Reflect

📝 Reflection Journal

After listening,

Gemini summarizes the user's reflection.

🧠 Memory Card

The next session remembers yesterday's experience.

RagaChakra becomes a companion instead of a playlist.

🧰 Tech Stack

Frontend

- React 18
- Vite
- React Router
- Framer Motion

Backend

- Node.js
- Express.js
- MongoDB
- Mongoose

AI

- Gemini 2.5 Flash

Utilities

- SunCalc
- JWT
- Helmet
- Express Validator
- Express Rate Limit

📁 Project Structure

...

client/
components/
features/
context/
styles/

server/
controllers/
services/
middleware/
routes/

...

📸 Screenshots

(Add your screenshots here)

- Recommendation Experience
- Ritual View
- AI Thinking
- Reflection
- Dashboard

🧩 Demo

Video:

(Link)

Live Demo:

(Link)

⚙️ Local Setup

```bash

git clone <https://github.com/VibhuSuneja/personalmusic>

```
cd personalmusic
```

```
npm install
```

```
...
```

```
Create
```

```
...
```

```
server/.env
```

```
...
```

```
...
```

```
MONGO_URI=
```

```
GEMINI_API_KEY=
```

```
JWT_SECRET=
```

```
...
```

```
Run
```

```
...
```

```
npm run dev
```

```
...
```

```

```

```
🌟 Future Roadmap
```

- Voice-guided rituals
- Personalized adaptive learning
- Raga progression plans
- Wearable integration
- Native mobile app

```

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🧑 Author
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Vibhu Suneja
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LinkedIn:
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https://linkedin.com/in/vibhusuneja08
```

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GitHub:
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https://github.com/VibhuSuneja
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★ Why RagaChakra?
```

```
> Music recommendation platforms optimize for attention.
```

## 3-Minute Demo Script

### Slide 1 (0–20 sec)

Hello everyone.

I'm Vibhu, and this is **RagaChakra**.

RagaChakra is an AI Ritual Companion that combines Indian Classical Music, traditional musicology, and Generative AI to create personalized listening rituals.

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### Problem (20–45 sec)

"Today's recommendation engines are designed to maximize engagement.

They recommend what's popular.

They don't consider

- how you're feeling,
- what time it is,
- or centuries of musical knowledge.

Indian Classical Music already has a framework for matching ragas to different times and emotional states.

That knowledge is difficult for modern listeners to access."

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### Solution (45–75 sec)

"So we built RagaChakra.

Instead of simply recommending another song,

it creates an intentional listening ritual.

The AI understands your mood,

combines it with the current Prahar,

filters culturally appropriate ragas,



and then generates an explainable recommendation."

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## **Architecture (75–105 sec)**

"Our architecture is hybrid.

The Rule Engine determines the correct Prahar.

MongoDB stores validated musical knowledge.

Gemini never invents ragas.

Instead, it chooses from culturally correct candidates and creates the explanation and ritual.

This keeps the system explainable and reduces hallucinations."

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## **Live Demo (105–165 sec)**

"I'll demonstrate the 'Overwhelmed Before Presentation' scenario.

1. Enter mood: Overwhelmed.
  2. Watch the AI Thinking animation.
  3. Receive Raga Yaman with a confidence score.
  4. Open the Explainability Card.
  5. Play the embedded audio.
  6. Begin the Ritual.
  7. Skip to Reflection using Demo Mode.
  8. Enter a short reflection.
  9. Show the AI-generated summary.
  10. Return to the dashboard and show the Memory Card with yesterday's insight."
- 

## **Impact (165–180 sec)**

"RagaChakra is not another music recommender.

It is an AI companion that helps users slow down, reflect, and reconnect with one of the world's richest musical traditions.

Thank you."

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## Expected Judge Questions & Answers

### Q: Why use AI instead of hardcoded rules?

A: The rules determine *which* ragas are culturally appropriate. AI personalizes *why* a particular raga fits the user's emotional context and guides the ritual in natural language.

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### Q: Why not let AI choose any raga?

A: We intentionally use a hybrid architecture. The rule engine enforces traditional constraints, while Gemini handles personalization and explanation. This improves reliability and reduces hallucinations.

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### Q: What makes this different from Spotify or YouTube?

A: Spotify optimizes for engagement based on listening history. RagaChakra optimizes for intentional listening using time, mood, and classical music theory to create a guided experience.

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### Q: How can this scale?

A: The architecture can expand with more ragas, richer metadata, multilingual support, wearable integrations, and partnerships with music platforms, while preserving the same hybrid recommendation pipeline.

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My final advice: **record the demo as if you're telling one story**. Avoid jumping between pages to show every feature. Judges are much more likely to remember a smooth narrative—from mood to recommendation to ritual to reflection—than a checklist of capabilities.